

CPSC368 Group 2 Project Phase 5: Formal Report

Introduction

Brief background

The film industry involves complex interactions between individual talent, collaboration, and representation. Movies are big projects which involve a large number of individuals and companies collaborating together and working closely in roughly the same standard procedure (Chisholm et al., 2015, as cited in Zhang et al., 2019). However, there is insignificant quantitative evidence of showing how interactions between stakeholders impact performance outcomes, particularly in movie ratings and career progression.

This has gained our interest to study how structural and creative factors in filmmaking influence performance outcomes, specifically in movie ratings and career progression. It moves beyond subjective opinion and explores whether measurable relationships exist between industry structure and success indicators. We aim to evaluate whether commonly held beliefs in the entertainment industry are supported by empirical evidence.

Our overarching theme is how characteristics of film production teams influence movie ratings and career outcomes across different genres.

Research questions

We have divided the big research question into three sub-questions we are going to investigate:

- What is the relationship between actors' and actresses' appearance in highly-rated movies and the later success of actors and actresses?
- What is the effect of repeated collaboration between directors and writers on the film's average rating, across action, romance and comedy genres, and fans' expectations?
- How does the average rating of movies vary based on the gender ratio of their principal cast across action, romance and comedy genres?

Who would care?

As movie industry involves complex interactions, it involves many stakeholders such as:

- Film studios and producers: In making strategic decisions about casting and creative partnerships.
- Casting directors: In evaluating talent and potential.
- Investors: In assessing risk and predictors of quality.
- Aspiring actors and filmmakers: In understanding what factors influence career advancement.

They would care about the answer to our research questions as these questions are highly related to these stakeholders.

Data

Datasets Used:

1. IMDb Non-Commercial Datasets

- Source: <https://developer.imdb.com/non-commercial-datasets/>
- Files Used: name.basics (actor names/professions), title.basics (titles, start year, genres, title types), title.crew (directors and writers), title.principals (job categories like 'actor'/'actress'), and title.ratings (average rating and vote counts).

2. MovieLens (ML) Dataset

- o Source: <https://grouplens.org/datasets/movielens/>
- o Files Used: links.csv, movies.csv, and ratings.csv.

3. TMDb Top 10,000 Popular Movies Dataset

- o Source: <https://www.kaggle.com/datasets/balaka18/tmdb-top-10000-popular-movies-dataset>
- o Files Used: movies_tmdb_popular.csv

Rationale for Dataset Selection

We chose these specific datasets because they each offer complementary attributes required to answer our overarching research question about film production characteristics and outcomes.

- **IMDb** is the main source for our data analysis as it provides comprehensive metadata on film titles, cast members, crew relationships, genres, release years, and critical reception (ratings). It contains millions of records and includes separate files for titles, crew, ratings, and cast members. These specific variables are needed to answer all three of our research questions, allowing us to analyze career trajectories, creative collaborations, and gender composition in relation to film ratings.
- **MovieLens** was crucial because it contains user ratings, tags, and most importantly, a links.csv file acting as a "Rosetta Stone." This linking file allowed us to seamlessly join internal MovieLens IDs with IMDb and TMDb identifiers. By linking these IDs, we were able to validate movie information across databases and join all three sources into a single, comprehensive dataset.
- **TMDb** contains the top 10,000 popular movies with popularity metrics. We selected it because it supports the analysis of top-performing movies when evaluating rating trends and genre distributions. Specifically, it provides two important variables: id (the TMDb ID used to identify movies) and popularity (a specific metric used to help identify the "Top 50" movies alongside IMDb vote counts).

Data Cleaning Steps

Given the massive scale of these databases, we performed several rigorous data cleaning and standardization steps to prepare the data for Oracle and MongoDB:

1. **Memory Management:** For the massive IMDb datasets, we initially loaded only the first 50,000 rows to prevent overloading our system RAM.
2. **Handling Missing Values:** Across all three datasets, we dropped any rows containing null or missing values to ensure query stability (especially to prevent downstream errors like division-by-zero during gender ratio calculations).

3. **ID Standardization:** MovieLens imdbId values are purely numeric, while native IMDb tconst IDs use a "tt" prefix. We appended the "tt" prefix to the MovieLens IDs to enable proper table joins.
4. **Format Standardization:** We converted all string data (genres, title types, job categories, and column names) to lowercase. Furthermore, MovieLens genres are separated by pipes (|), which we converted into comma-separated arrays to match the IMDb genre format.
5. **Row Truncation:** To ensure our queries ran efficiently across both database systems for the scope of this project, we randomly sampled and filtered the final combined dataset down to exactly 2,000 rows.

Exploratory Data Analysis (EDA) Discoveries

During our initial EDA, we discovered several structural patterns and anomalies:

- **Format Inconsistencies:** We discovered early on that genres were highly disorganized. A single movie might be tagged as ["Action", "Comedy"] in IMDb but Action|Comedy in MovieLens. This necessitated our pipeline's standardization step and the use of explicit CASE and regex-style matching (LIKE '%ACTION%') in our SQL queries.
- **Missing Data:** We noticed that older, obscure, or non-Western short films frequently lacked complete crew data or had empty director / writer fields (often populated as \N in IMDb).
- **Outliers & Skewed Distributions:** When exploring cast demographics, we found that balanced casts were surprisingly rare compared to heavily male-dominated casts. Furthermore, female-dominated Action movies represented an extreme minority in our dataset, acting as an outlier category that nonetheless scored exceptionally high in average ratings.

Important Limitations and Biases

- **Selection and Survivorship Bias:** By dropping all rows containing null values, we likely introduced a survivorship bias. Modern, big-budget Western films are far more likely to have complete metadata on IMDb than older, independent, or international films. Consequently, our dataset is heavily biased toward mainstream modern cinema.
- **The 2,000-Row Constraint:** Capping our final dataset at 2,000 rows is the most severe limitation. This dramatically reduces our statistical power and means our sample is not fully representative of the millions of titles in the global film industry. Trends observed in our results may be affected by this small sample size rather than industry-wide truths.

Methodology

Analytical Strategy

The analysis was done by extracting variables from the database using query-based aggregation techniques. The goal was to transform raw movie metadata into measurable variables that could be used to answer our research questions and evaluate relationships between production characteristics and movie ratings.

Both Oracle and MongoDB databases were used to implement this analysis. The same analytical logic was applied in both systems to ensure the same resulting data, with these differences in approaches:

- Oracle: relied on SQL queries combining multiple tables using joins and Common Table Expressions (CTEs).
- MongoDB: used aggregation pipelines that transform and aggregate document data through stages.

Statistical Techniques

The analysis relied mainly on aggregation-based statistical summaries, including:

- Counts (e.g., number of roles per actor or number of director–writer collaborations)
- Averages (e.g., average IMDb rating of movies)
- Ratios (e.g., male-to-total actor ratio in a movie cast)
- Categorical grouping (e.g., grouping films into Action, Romance, or Comedy genres)

These metrics were constructed and used for visualizations that explain the relationships being evaluated. The results will be further discussed in the results section.

Application to Research Questions

The methodology was applied to each research question as follows.

- Actor success analysis: Actors were grouped by gender and genre, and the number of roles and average rating of their films were calculated.
- Collaboration analysis: Director–writer pairs were identified and the number of collaborations was linked with the rating of their films.
- Gender representation analysis: The ratio of male actors to total lead actors was calculated for each movie and classified as male-dominated, female-dominated, or balanced before comparing ratings.

Rationale for the Approach

This approach was chosen because the research questions require identifying relationships between multiple entities such as actors, creative teams, and movies. Aggregation queries efficiently compute the counts, averages, and ratios needed for this analysis, while allowing the same logic to be implemented in both Oracle and MongoDB for comparison.

Database Implementation

Oracle Implementation

In the Oracle implementation, each cleaned dataset was stored as a separate relational table corresponding to the original CSV files, including tables derived from the IMDb, MovieLens, and TMDb datasets.

Key entities such as movies, cast members, crew information, and ratings were therefore maintained in different tables. Relationships between these tables were established through shared identifiers such as movie IDs and person IDs, which allowed the data to be combined when necessary using SQL joins during analysis.

MongoDB Document Implementation

In the MongoDB implementation, the data was stored using a document-oriented design to reduce the need for joins. The collections were:

- Movies collection – stores movie-level information such as title, year, genres, ratings, popularity, crew, and cast.
- People collection – stores individuals and their filmography.
- UserRatings collection – stores MovieLens user rating records.

The Movies collection embeds related data, such as cast members and crew, directly inside each movie document. For example:

- crew contains arrays of director and writer IDs
- cast contains embedded documents describing actors appearing in the film

Embedding was chosen because these attributes are closely associated with a movie and are frequently accessed together during analysis.

Design Decisions

The main design decision in the MongoDB implementation was deciding when to embed documents versus store data in separate collections.

- Embedded structures were used for cast and crew information because they are naturally tied to a movie and are typically queried together with the movie document.
- Separate collections were used for entities such as people and user ratings because these records can be associated with multiple movies and may also be queried independently.

For this dataset, the **MongoDB document model felt more natural** because movie-related attributes such as cast, crew, genres, and ratings could be stored together within a single document. This structure simplifies many analytical queries by reducing the need to combine multiple tables. The comparison between these two implementations will be further discussed in the next section.

Database Comparison and Reflection

Preferred Database Overall

For this project, MongoDB would be the preferred database. The document-based structure allows movie-related information such as genres, cast members, crew, and ratings to be stored together within a single document. This structure aligns naturally with the way movie data is organized and simplifies many analytical queries because the data does not need to be reconstructed from multiple tables.

In contrast, the Oracle implementation required combining information from several relational tables using joins before performing the analysis.

When Oracle Would be Preferred

Oracle would become the better choice if the project required stronger relational structure and data integrity across entities. For example, Oracle would be more appropriate if:

- The project focused more heavily on relationships between multiple entities, such as actors, directors, and movies across many tables.
- The analysis required frequent joins across datasets rather than retrieving information centered around a single movie.
- The database needed strict constraints and normalization, such as enforcing foreign key relationships between people, movies, and roles.

If the project required more complex relational queries across separate entities rather than document-style movie records, the relational structure in Oracle would be more efficient.

Query Complexity across Databases

For the research questions in this project:

- Oracle queries required combining several normalized tables using joins and CTEs before performing aggregations.
- MongoDB queries used aggregation pipelines that directly processed embedded arrays and documents.

Because cast and crew information were embedded within movie documents, MongoDB queries often avoided the need for multiple joins.

Queries That Were Easier in One System

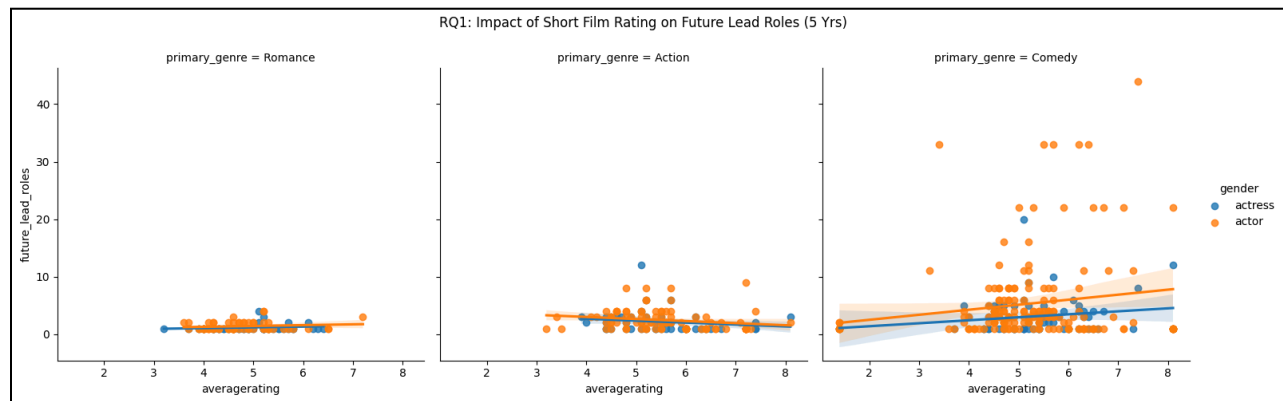
- Cast-based analyses (such as calculating gender ratios or counting actor roles) were easier in MongoDB because the cast data was stored as an embedded array that could be expanded using \$unwind.
- Relational joins between datasets were easier in Oracle because SQL provides a clear structure for combining normalized tables.

Overall, MongoDB simplified queries that involved nested movie data, while Oracle provided clearer support for combining multiple relational tables.

Results

Research Question 1: The relationship between appearance in highly-rated movies & their later success.

Initial Hypothesis: We hypothesized that actors and actresses who appear in highly-rated movies will experience greater subsequent career success, measured by a higher volume of future lead roles within the next 5 years.



Visualization Interpretation: The visualization below consists of three scatter plots, faceted by the primary genre of the initial film (Romance, Action, and Comedy). The x-axis represents the average IMDb rating of the initial short or feature film, and the y-axis represents the number of future lead roles the actor/actress secured. The data points are color-coded by gender (blue for actresses, orange for actors), and linear regression lines are applied to visualize the general trend.

Findings and Discussion: Our results show a mixed conclusion that partially contradicts our initial hypothesis, as the correlation is genre dependent.

- **Romance and Action:** For both the Romance and Action genres, the regression lines are almost flat. This indicates that appearing in a highly-rated Action or Romance movie has almost no noticeable impact on the raw volume of future lead roles an actor or actress receives. Success in these genres appears to be isolated, or driven by factors outside of pure IMDb ratings (such as box office revenue or studio contracts).
- **Comedy:** The Comedy genre is the only category that supports our hypothesis. There is a positive correlation where higher average ratings correspond to an increase in future lead roles. Interestingly, the data also highlights a gender disparity in this genre, male actors (orange) show a steeper upward trend and reach much higher peaks in future lead roles (upwards of 30-40 roles) compared to actresses when starring in highly-rated comedies.

The data suggests that success (high ratings) only reliably translates to career success (more roles) within the Comedy genre.

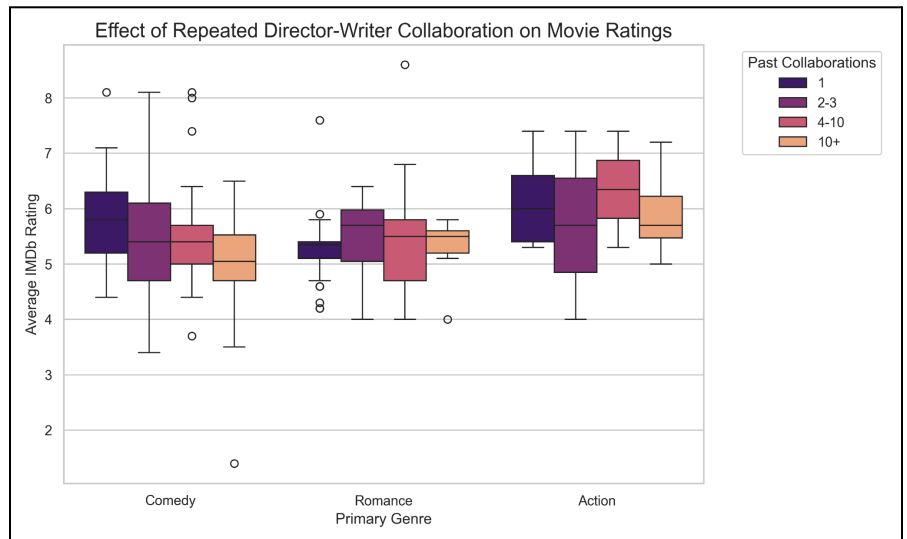
Research Question 2: The effect of repeated collaboration on rating and fans' expectations.

Initial Hypothesis: We hypothesized that a higher number of past collaborations between a specific director and writer would lead to higher average IMDb ratings, building on the assumption that repeated collaboration yields a chemistry and refined storytelling.

Visualization Explanation: The box plot below illustrates the distribution of average IMDb ratings (y-axis) based on the frequency of past collaborations between a director and writer (grouped into buckets: 1, 2-3, 4-10, and 10+ collaborations). The data is further segmented by primary genre (Comedy, Romance, Action) on the x-axis to observe if ratings vary by movie type.

Findings and Discussion: The results contradict our initial hypothesis. Rather than improving movie quality, repeated collaborations generally correlate with stagnant or declining ratings.

- **Comedy:** As the number of collaborations increases from 1 to 10+, both the median rating and the overall distribution steadily decrease. This suggests that in comedy, repeated partnerships might lead to creative stagnation or audience fatigue.
- **Romance:** The trend in Romance is similarly uninspiring. The median rating remains flat regardless of whether the duo has worked together once or over ten times.
- **Action:** The Action genre presents a bell-curve trend. The median rating dips for 2-3 collaborations but hits a peak at the 4-10 collaboration mark, before dropping significantly for teams with 10+ collaborations. This implies there is a sweet spot for Action production teams where they hit their creative stride, but eventually, the partnership results in diminishing returns.



Overall, the data indicates that fresh director-writer pairings (1 collaboration) often produce higher-rated films than established teams (10+ collaborations).

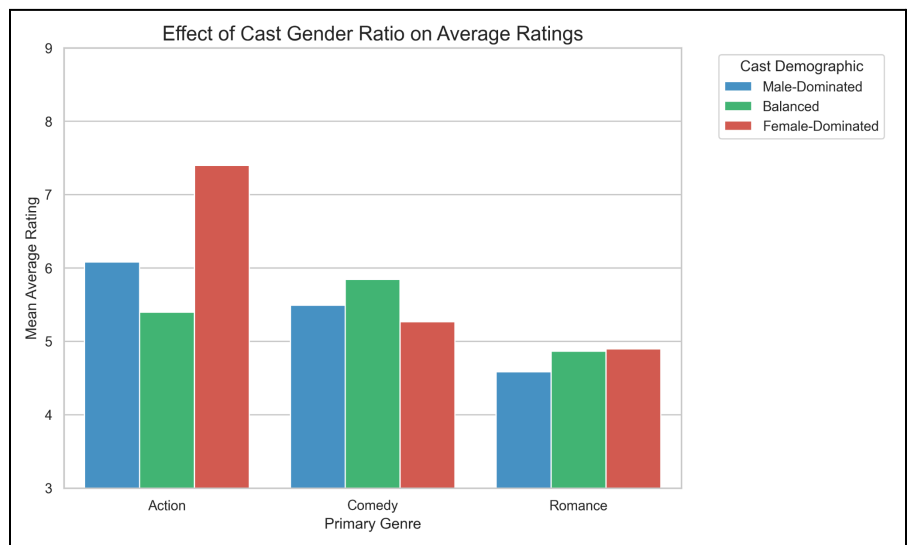
Research Question 3: Variation of average rating based on gender ratio in the principal cast.

Initial Hypothesis: We hypothesized that movies with a balanced gender ratio in their principal cast would achieve the highest average ratings across all genres, as balanced representation would theoretically appeal to a larger audience demographic.

Visualization Explanation: The grouped bar chart below displays the mean average IMDb rating (y-axis) categorized by the principal cast's demographic makeup (Male-Dominated, Balanced, Female-Dominated). The results are separated into Action, Comedy, and Romance genres along the x-axis to track how audience reception to gender ratios shifts depending on the film's subject matter.

Findings and Discussion: The results present a picture that only partially supports our hypothesis, showing that audience preferences for cast demographics are tied to genre expectations.

- **Comedy:** The results in the Comedy genre support our hypothesis. Balanced casts yield the highest mean average rating (~5.8), outperforming both Male-Dominated and Female-Dominated casts.
- **Romance:** In the Romance genre, Female-Dominated and Balanced casts perform almost equally well (~4.9), both noticeably outperforming Male-Dominated casts (~4.5). This partially aligns with our hypothesis but indicates a slight genre bias where female-led casts are accepted.
- **Action:** The Action genre entirely contradicts our hypothesis. Female-Dominated casts achieved a higher mean average rating (~7.4) than both Male-Dominated (~6.0) and Balanced (~5.4) casts. While traditional industry stereotypes might suggest action is a male-dominated space, this data shows that when female-dominated action movies are made, they are rated exceptionally high by audiences.
- In conclusion, a balanced cast is optimal for general genres like Comedy, but audiences show distinct, skewed preferences in targeted genres like Action and Romance.



Limitations and Future Work

Limitations of the Study:

- **Data and Scope Restrictions:** The most significant limitation of our study is the restriction of our dataset size. To manage computational constraints and avoid overloading RAM, we down-sampled our IMDb, MovieLens, and TMDb datasets to exactly 2,000 rows. While this allowed us to successfully model our Oracle and MongoDB schemas, a 2,000-row subset is not representative of the millions of titles in the global film industry. Consequently, our findings may be prone to small-sample-size bias and might not generalize to the broader cinematic landscape.
- **Threats to Validity:** In Research Question 1, a threat to validity exists within our methodology for tracking future success. Because our SQL/MongoDB queries did not include a chronological date filter (e.g., `future_movie.startyear > initial_movie.startyear`), we correlated a specific movie's rating with the actor's *total career volume* in that genre rather than subsequent roles. Therefore, we cannot definitively prove that a highly-rated short film *caused* the future roles; we only established a correlation between actors who appear in highly-rated movies and actors who have a high volume of overall roles.
- **Omitted Variables Bias:** Our analysis relies heavily on IMDb ratings and demographic/crew characteristics. However, we were unable to factor in external economic drivers such as movie budgets, marketing spend, or franchise IP (e.g., Marvel or Star Wars). A high-budget marketing campaign might drive up the popularity and rating of an action movie far more than the gender ratio of its cast, meaning our current correlations might be masking stronger variables.

Future Work: If given more time and computing resources, we would prioritize scaling this analysis to the complete, un-truncated IMDb and MovieLens datasets using a distributed cloud database system. Analyzing millions of rows would provide the statistical power needed to confirm whether the trends we observed hold true at a larger scale. Furthermore, we would integrate financial datasets to evaluate whether box office revenue and production budget are stronger predictors of an actor's future casting success than ratings.

Uninvestigated Questions Raised by This Work

This research highlighted several unexpected trends that warrant deeper investigation in the future:

1. **The Female-Dominated Action Spike:** Why did female-dominated action movies score so disproportionately high compared to male-dominated ones? Is this a genuine shift in modern audience preferences, or simply a result of a very small sample size of female-led action films within our 2,000-row subset?
2. **The Comedy Collaboration Penalty:** Why does repeated collaboration seemingly hurt movie ratings exclusively in the comedy genre? Does comedic success rely more heavily on novelty and fresh perspectives than romance or action?

References

- IMDb. (n.d.). *IMDb non-commercial datasets*. IMDb Developer.
<https://developer.imdb.com/non-commercial-datasets/>
- GroupLens Research. (2023). *MovieLens 32M dataset*. GroupLens Research, University of Minnesota.
<https://grouplens.org/datasets/movielens/>
- Balaka18. (n.d.). *TMDb Top 10,000 popular movies dataset*. Kaggle.
<https://www.kaggle.com/datasets/balaka18/tmdb-top-10000-popular-movies-dataset>

AI Tool Use Declaration

- We used Gemini to check for formatting and grammatical mistakes for this document.
<https://gemini.google.com/share/ddd54a870b85>
- We used ChatGPT in a previous phase to help with formatting and checking issues with our methodologies to combine the datasets.
<https://chatgpt.com/share/69a373a1-80e4-8009-81ab-1816ff137ddf>
- We used Gemini to address the limitations of our work
<https://gemini.google.com/share/69299cac97a0>